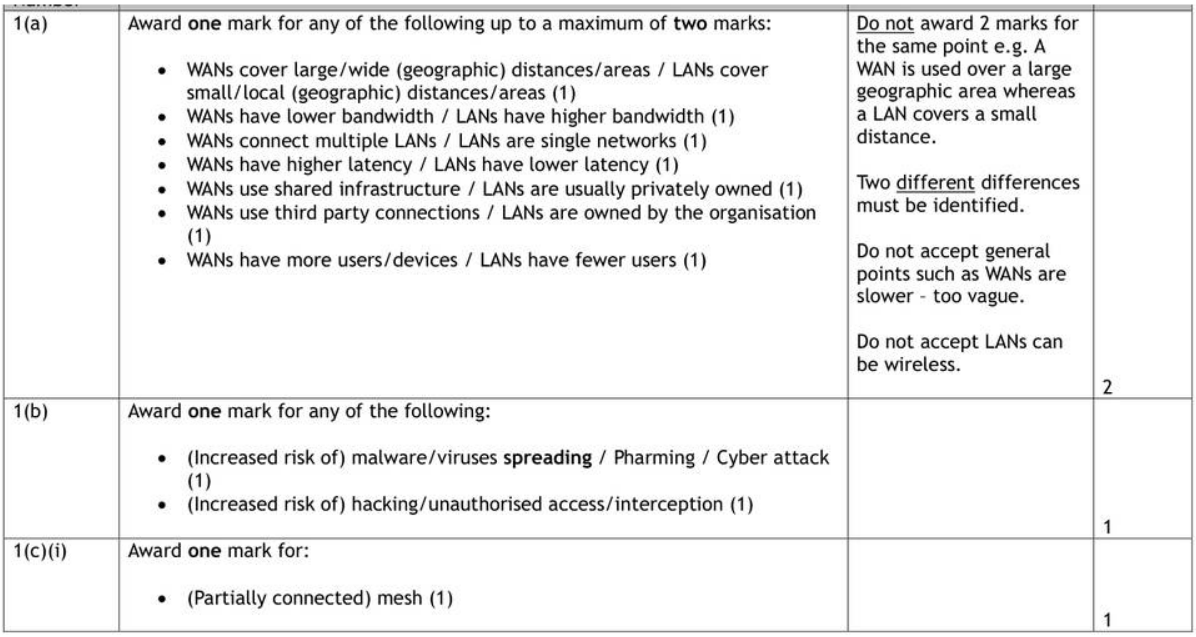
Type MS · Network types: LAN, WAN, PAN and why we network

Answer 1 of 10 · Question 1 part a · 2 marks

2024 JUN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

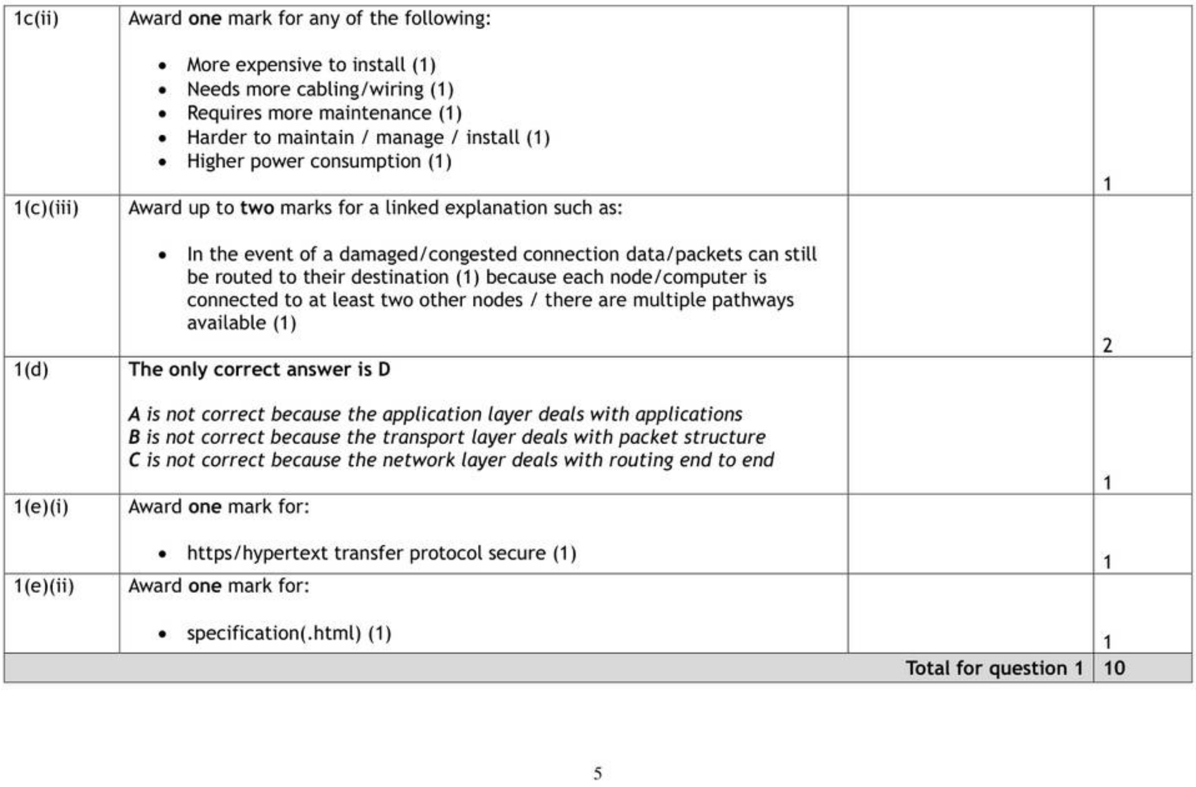


1(a)	Award **one** mark for any of the following up to a maximum of **two** marks: • WANs cover large/wide (geographic) distances/areas / LANs cover small/local (geographic) distances/areas (1) • WANs have lower bandwidth / LANs have higher bandwidth (1) • WANs connect multiple LANs / LANs are single networks (1) • WANs have higher latency / LANs have lower latency (1) • WANs use shared infrastructure / LANs are usually privately owned (1) • WANs use third party connections / LANs are owned by the organisation (1) • WANs have more users/devices / LANs have fewer users (1)	Do not award 2 marks for the same point e.g. A WAN is used over a large geographic area whereas a LAN covers a small distance. Two different differences must be identified. Do not accept general points such as WANs are slower - too vague. Do not accept LANs can be wireless.	2
1(b)	Award **one** mark for any of the following: • (Increased risk of) malware/viruses **spreading** / Pharming / Cyber attack (1) • (Increased risk of) hacking/unauthorised access/interception (1)		1
1(c)(i)	Award **one** mark for: • (Partially connected) mesh (1)		1

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 1 of 10 · Question 1 part a · 2 marks

2024 JUN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract



1c(ii)	Award **one** mark for any of the following: • More expensive to install (1) • Needs more cabling/wiring (1) • Requires more maintenance (1) • Harder to maintain / manage / install (1) • Higher power consumption (1)		1
1(c)(iii)	Award up to **two** marks for a linked explanation such as: • In the event of a damaged/congested connection data/packets can still be routed to their destination (1) because each node/computer is connected to at least two other nodes / there are multiple pathways available (1)		2
1(d)	**The only correct answer is D** ***A*** *is not correct because the application layer deals with applications* ***B*** *is not correct because the transport layer deals with packet structure* ***C*** *is not correct because the network layer deals with routing end to end*		1
1(e)(i)	Award **one** mark for: • https/hypertext transfer protocol secure (1)		1
1(e)(ii)	Award **one** mark for: • specification(.html) (1)		1
		Total for question 1	**10**

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 1 of 10 · Question 1 part a · 2 marks

2024 JUN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 2 of 10 · Question 3 part a.i · 1 mark

2019 JUN · Paper 01 · extract 1 of 4 · find part a.i in this mark scheme extract

3(a)(i)	• LAN / Local area network (1) Allow PAN/Personal Area Network		1

Question Number	Answer	Additional Guidance	Mark
3(a)(ii)	• WAN / Wide area network (1)		1

Question Number	Answer	Additional Guidance	Mark
3(b)	A: Megabits per second **The only correct answer is A** **B** is not correct because megapixels are associated with images rather than network data speeds **C** is not correct because mebicycles are not associated with network data speeds **D** is not correct because mebihertz are not associated with network data speeds		1

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 2 of 10 · Question 3 part a.i · 1 mark

2019 JUN · Paper 01 · extract 2 of 4 · find part a.i in this mark scheme extract

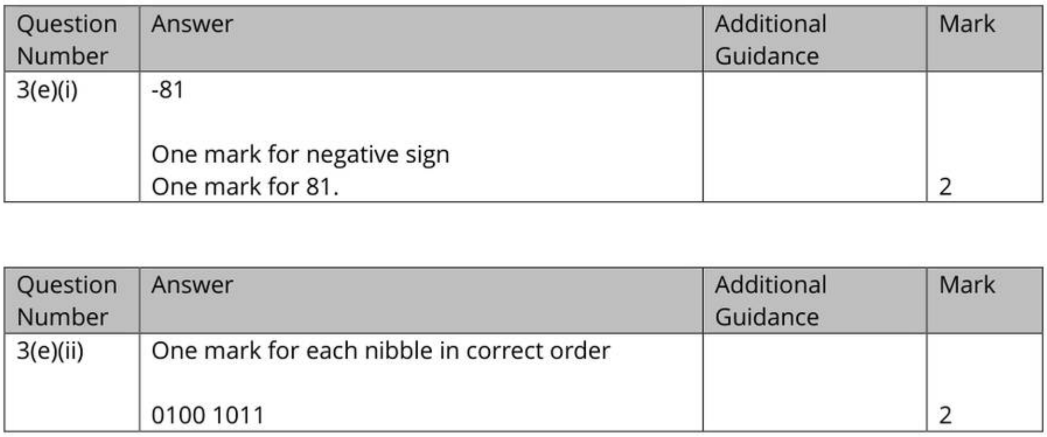
Question Number	Answer	Additional Guidance	Mark
3(c)	• IMAP (1) AND **one** from: • because changes are synced (1) with the mail server in real-time (1) • because the messages always reside on the email server (1) as she has limited storage (1) on her tablet and phone • it gets sent to both smartphone and tablet (1) **Alternative answer:** • POP3 (not awarded a mark) AND **one** from: • because storage limitations on server (1) and privacy issues (1) • because more CPU time (1) may be used when syncing (1) (a large number of messages).		3

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(d)</td><td>
<table>
<tr><th>Characteristic</th><th>Bus</th><th>Ring</th><th>Star</th><th>Mesh</th></tr>
<tr><td>Network performance degrades as more devices are added.</td><td>✓</td><td></td><td></td><td></td></tr>
<tr><td>All devices are directly or indirectly connected to all others.</td><td></td><td></td><td></td><td>✓</td></tr>
<tr><td>Each node has a physical attachment to a routing device.</td><td></td><td></td><td>✓</td><td></td></tr>
<tr><td>There are no data collisions, because packets travel in the same direction.</td><td></td><td>✓</td><td></td><td></td></tr>
</table>
</td><td></td><td>4</td></tr>
</table>

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 2 of 10 · Question 3 part a.i · 1 mark

2019 JUN · Paper 01 · extract 3 of 4 · find part a.i in this mark scheme extract



Question Number	Answer	Additional Guidance	Mark
3(e)(i)	-81 One mark for negative sign One mark for 81.		2

Question Number	Answer	Additional Guidance	Mark
3(e)(ii)	One mark for each nibble in correct order 0100 1011		2

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 2 of 10 · Question 3 part a.i · 1 mark

2019 JUN · Paper 01 · extract 4 of 4 · find part a.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(e)(iii)	Any **two** from: 1. This is an overflow (1) error 2. It is caused by adding two 1s in the most significant bit (1) 3. The resulting number is too large to fit in the number of bits available / requires 9 bits / the number is larger than 2^8 / there is 1 that needs to be carried over (in MSB) / MSB gets discarded (1) 4. It can cause the program to crash or produce unreliable/incorrect results (1)		2

Question Number	Answer	Additional Guidance	Mark

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 3 of 10 · Question 3 part a.ii · 1 mark

2019 JUN · Paper 01 · extract 1 of 4 · find part a.ii in this mark scheme extract

3(a)(i)	- LAN / Local area network (1) Allow PAN/Personal Area Network		1

Question Number	Answer	Additional Guidance	Mark
3(a)(ii)	- WAN / Wide area network (1)		1

Question Number	Answer	Additional Guidance	Mark
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Type MS · Network types: LAN, WAN, PAN and why we network

Answer 3 of 10 · Question 3 part a.ii · 1 mark

2019 JUN · Paper 01 · extract 2 of 4 · find part a.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(c)	• IMAP (1) AND **one** from: • because changes are synced (1) with the mail server in real-time (1) • because the messages always reside on the email server (1) as she has limited storage (1) on her tablet and phone • it gets sent to both smartphone and tablet (1) **Alternative answer:** • POP3 (not awarded a mark) AND **one** from: • because storage limitations on server (1) and privacy issues (1) • because more CPU time (1) may be used when syncing (1) (a large number of messages).		3

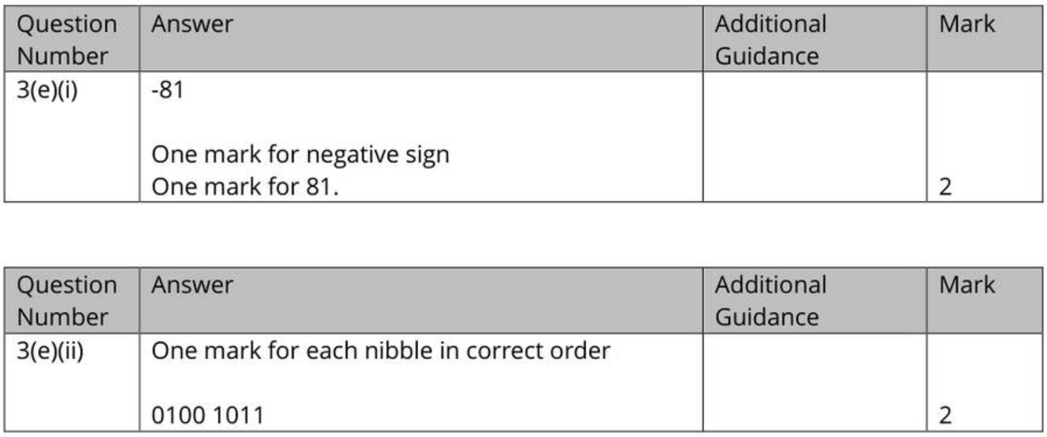
Question Number	Answer	Additional Guidance	Mark
3(d)	see table below		4

Characteristic	Bus	Ring	Star	Mesh
Network performance degrades as more devices are added.	✓			
All devices are directly or indirectly connected to all others.				✓
Each node has a physical attachment to a routing device.			✓	
There are no data collisions, because packets travel in the same direction.		✓		

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 3 of 10 · Question 3 part a.ii · 1 mark

2019 JUN · Paper 01 · extract 3 of 4 · find part a.ii in this mark scheme extract



Question Number	Answer	Additional Guidance	Mark
3(e)(i)	-81 One mark for negative sign One mark for 81.		2

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3(e)(ii)	One mark for each nibble in correct order 0100 1011		2

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 3 of 10 · Question 3 part a.ii · 1 mark

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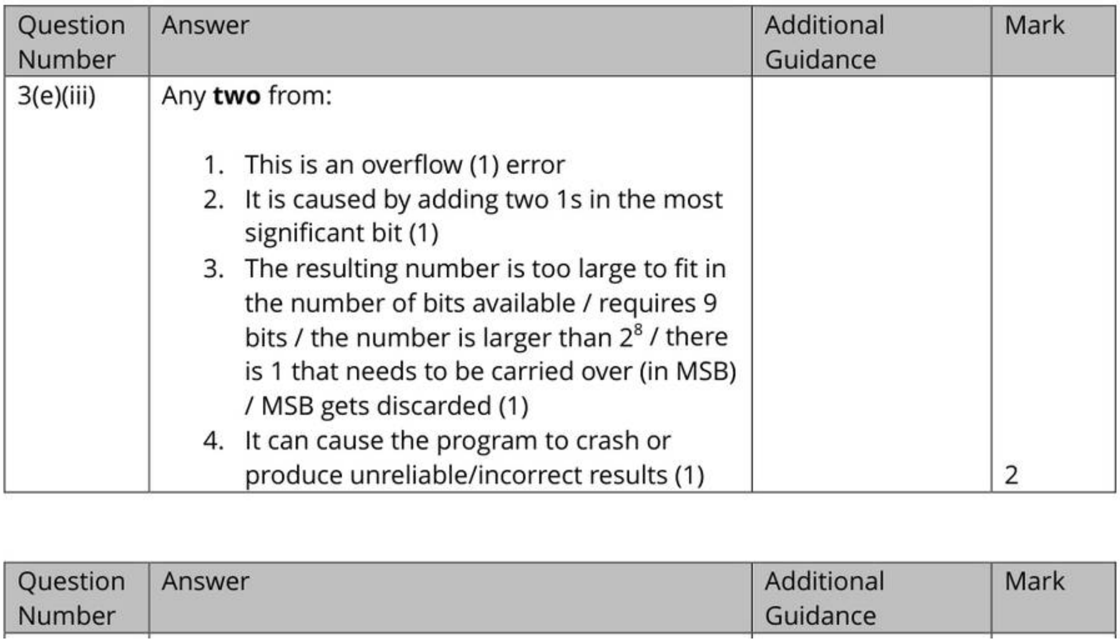


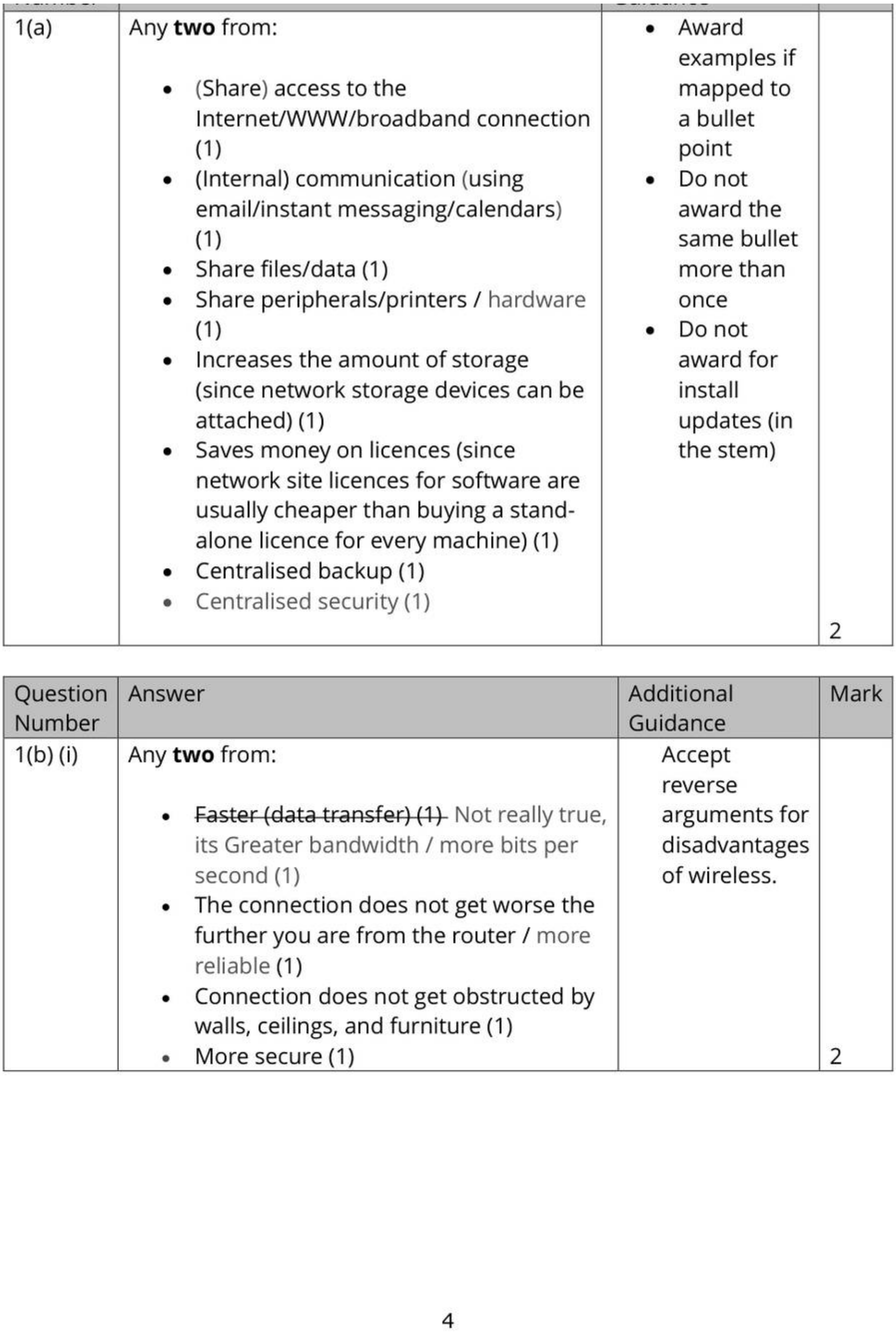
Question Number	Answer	Additional Guidance	Mark
3(e)(iii)	Any **two** from: 1. This is an overflow (1) error 2. It is caused by adding two 1s in the most significant bit (1) 3. The resulting number is too large to fit in the number of bits available / requires 9 bits / the number is larger than 2^8 / there is 1 that needs to be carried over (in MSB) / MSB gets discarded (1) 4. It can cause the program to crash or produce unreliable/incorrect results (1)		2

Question Number	Answer	Additional Guidance	Mark

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 4 of 10 · Question 1 part a · 2 marks

2020 NOV · Paper 01 · extract 1 of 5 · find part a in this mark scheme extract



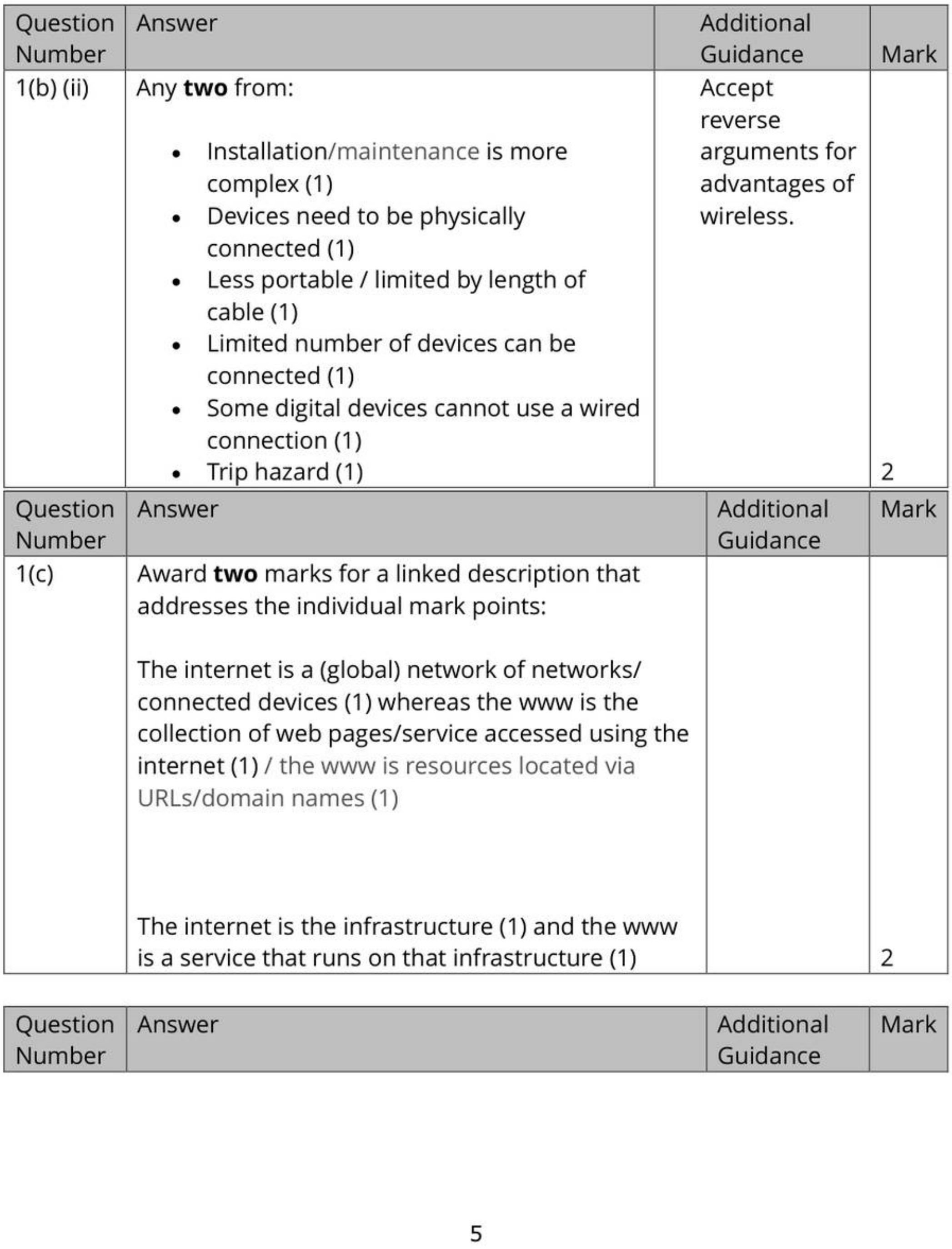
1(a)	Any **two** from: • (Share) access to the Internet/WWW/broadband connection (1) • (Internal) communication (using email/instant messaging/calendars) (1) • Share files/data (1) • Share peripherals/printers / hardware (1) • Increases the amount of storage (since network storage devices can be attached) (1) • Saves money on licences (since network site licences for software are usually cheaper than buying a stand-alone licence for every machine) (1) • Centralised backup (1) • Centralised security (1)	• Award examples if mapped to a bullet point • Do not award the same bullet more than once • Do not award for install updates (in the stem)	2

Question Number	Answer	Additional Guidance	Mark
1(b) (i)	Any **two** from: • ~~Faster (data transfer) (1)~~ Not really true, its Greater bandwidth / more bits per second (1) • The connection does not get worse the further you are from the router / more reliable (1) • Connection does not get obstructed by walls, ceilings, and furniture (1) • More secure (1)	Accept reverse arguments for disadvantages of wireless.	2

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 4 of 10 · Question 1 part a · 2 marks

2020 NOV · Paper 01 · extract 2 of 5 · find part a in this mark scheme extract



Question Number	Answer	Additional Guidance	Mark
1(b) (ii)	Any **two** from: • Installation/maintenance is more complex (1) • Devices need to be physically connected (1) • Less portable / limited by length of cable (1) • Limited number of devices can be connected (1) • Some digital devices cannot use a wired connection (1) • Trip hazard (1)	Accept reverse arguments for advantages of wireless.	2

Question Number	Answer	Additional Guidance	Mark
1(c)	Award **two** marks for a linked description that addresses the individual mark points: The internet is a (global) network of networks/ connected devices (1) whereas the www is the collection of web pages/service accessed using the internet (1) / the www is resources located via URLs/domain names (1) The internet is the infrastructure (1) and the www is a service that runs on that infrastructure (1)		2

Question Number	Answer	Additional Guidance	Mark

5

Answer 4 of 10 · Question 1 part a · 2 marks

2020 NOV · Paper 01 · extract 3 of 5 · find part a in this mark scheme extract

1(d)	Award **two** marks for a linked explanation such as: Running out of IPv4 addresses/the number of (internet) connected devices has grown (1) IPv6 addresses are long/longer than IPv4 / the number of possible addresses is large/will last much longer (1)		2

Question Number	Answer	Additional Guidance	Mark
1(e)(i)	Star		1

Question Number	Answer	Additional Guidance	Mark
1(e) (ii)	Any **one** from: • easy to connect/remove new nodes (1) • failure of one node/link does not affect the rest of the network (1) • easy to detect the failure of one node/link (1)		1

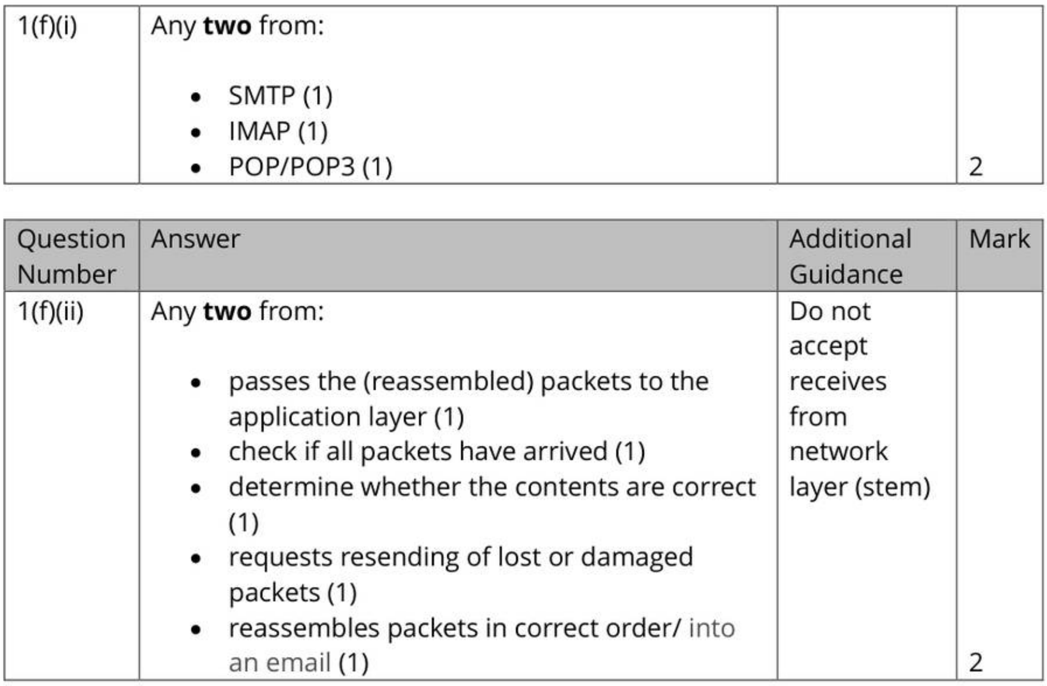
Question Number	Answer	Additional Guidance	Mark
1(e) (iii)	Any **one** from: • if central switch/hub fails then the whole network fails (1) • performance and number of nodes that can be added depends on capacity of central switch/hub (1) • can require more cable than some of the other topologies (1)		1

Question Number	Answer	Additional Guidance	Mark

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 4 of 10 · Question 1 part a · 2 marks

2020 NOV · Paper 01 · extract 4 of 5 · find part a in this mark scheme extract



1(f)(i)	Any **two** from: • SMTP (1) • IMAP (1) • POP/POP3 (1)		2

Question Number	Answer	Additional Guidance	Mark
1(f)(ii)	Any **two** from: • passes the (reassembled) packets to the application layer (1) • check if all packets have arrived (1) • determine whether the contents are correct (1) • requests resending of lost or damaged packets (1) • reassembles packets in correct order/ into an email (1)	Do not accept receives from network layer (stem)	2

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 4 of 10 · Question 1 part a · 2 marks

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Question Number	Answer	Additional Guidance	Mark

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 5 of 10 · Question 2 part a · 1 mark

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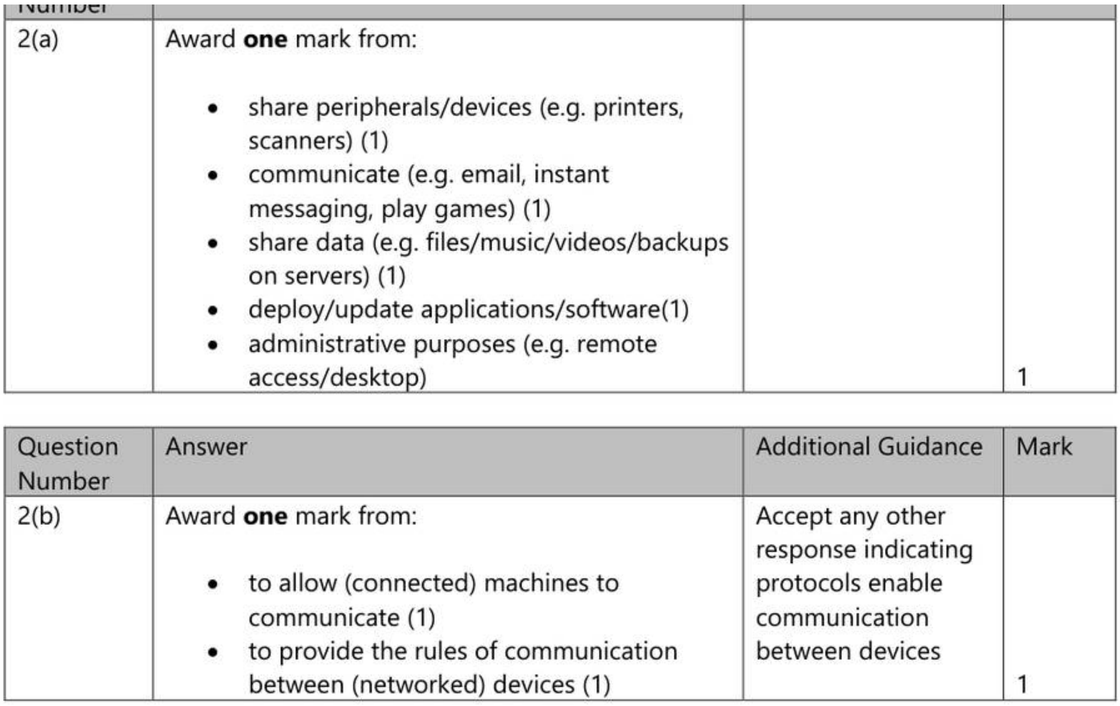


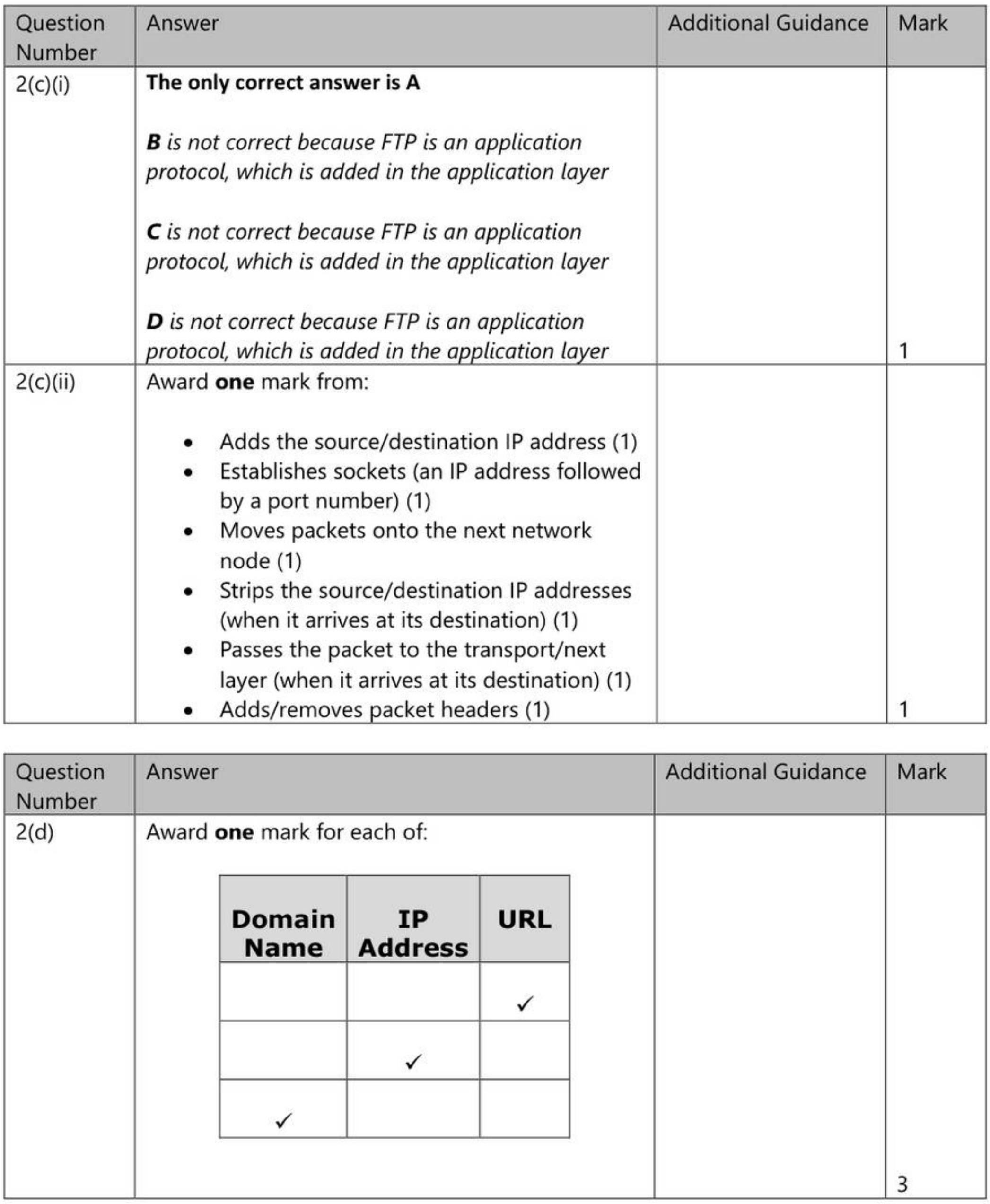
2(a)	Award **one** mark from: • share peripherals/devices (e.g. printers, scanners) (1) • communicate (e.g. email, instant messaging, play games) (1) • share data (e.g. files/music/videos/backups on servers) (1) • deploy/update applications/software(1) • administrative purposes (e.g. remote access/desktop)		1

Question Number	Answer	Additional Guidance	Mark
2(b)	Award **one** mark from: • to allow (connected) machines to communicate (1) • to provide the rules of communication between (networked) devices (1)	Accept any other response indicating protocols enable communication between devices	1

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 5 of 10 · Question 2 part a · 1 mark

2021 JUN · Paper 01 · extract 2 of 5 · find part a in this mark scheme extract



Question Number	Answer	Additional Guidance	Mark
2(c)(i)	**The only correct answer is A** ***B*** *is not correct because FTP is an application protocol, which is added in the application layer* ***C*** *is not correct because FTP is an application protocol, which is added in the application layer* ***D*** *is not correct because FTP is an application protocol, which is added in the application layer*		1
2(c)(ii)	Award **one** mark from: • Adds the source/destination IP address (1) • Establishes sockets (an IP address followed by a port number) (1) • Moves packets onto the next network node (1) • Strips the source/destination IP addresses (when it arrives at its destination) (1) • Passes the packet to the transport/next layer (when it arrives at its destination) (1) • Adds/removes packet headers (1)		1

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>2(d)</td><td>Award one mark for each of:
<table>
<tr><th>Domain Name</th><th>IP Address</th><th>URL</th></tr>
<tr><td></td><td></td><td>✓</td></tr>
<tr><td></td><td>✓</td><td></td></tr>
<tr><td>✓</td><td></td><td></td></tr>
</table>
</td><td></td><td>3</td></tr>
</table>

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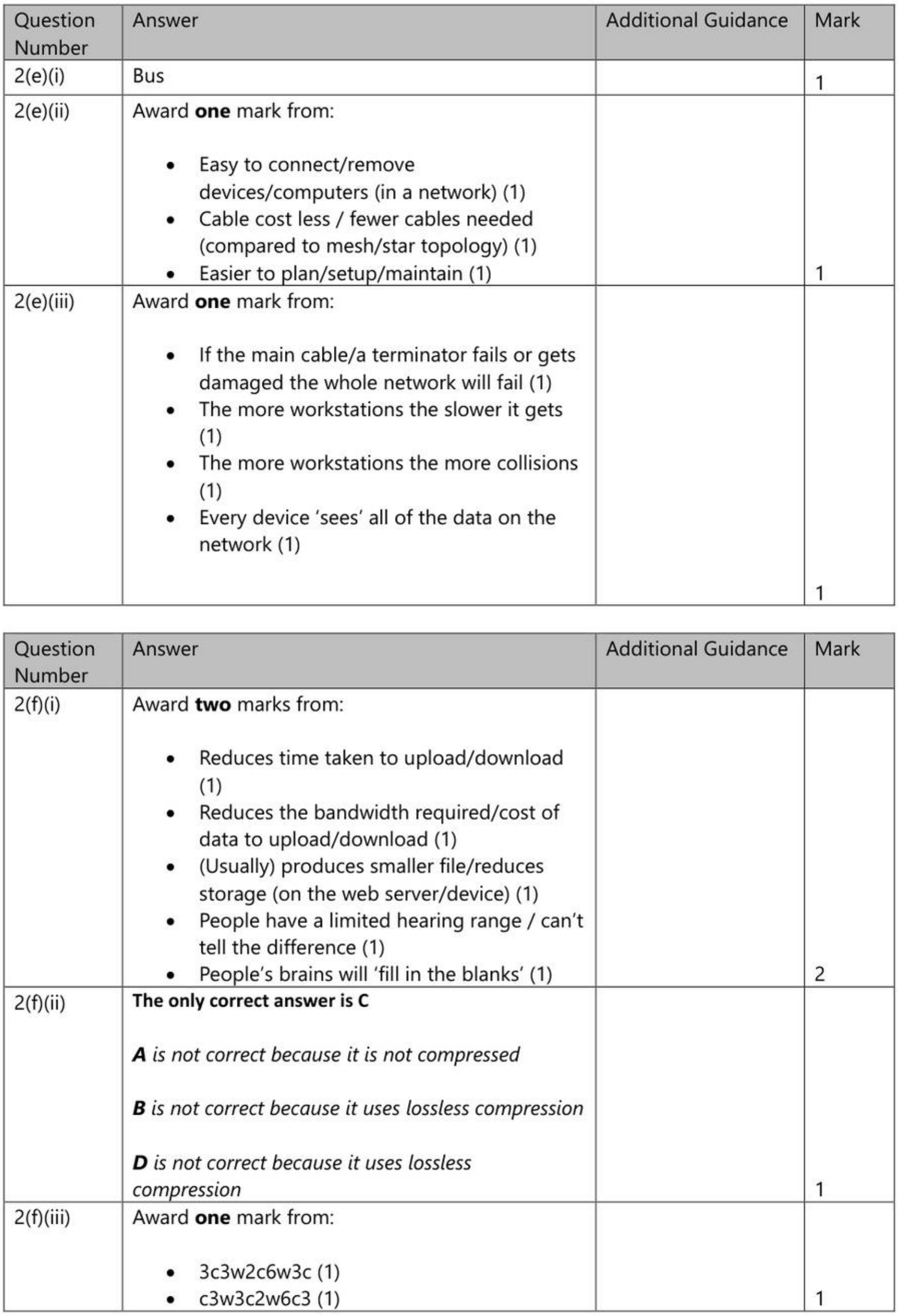


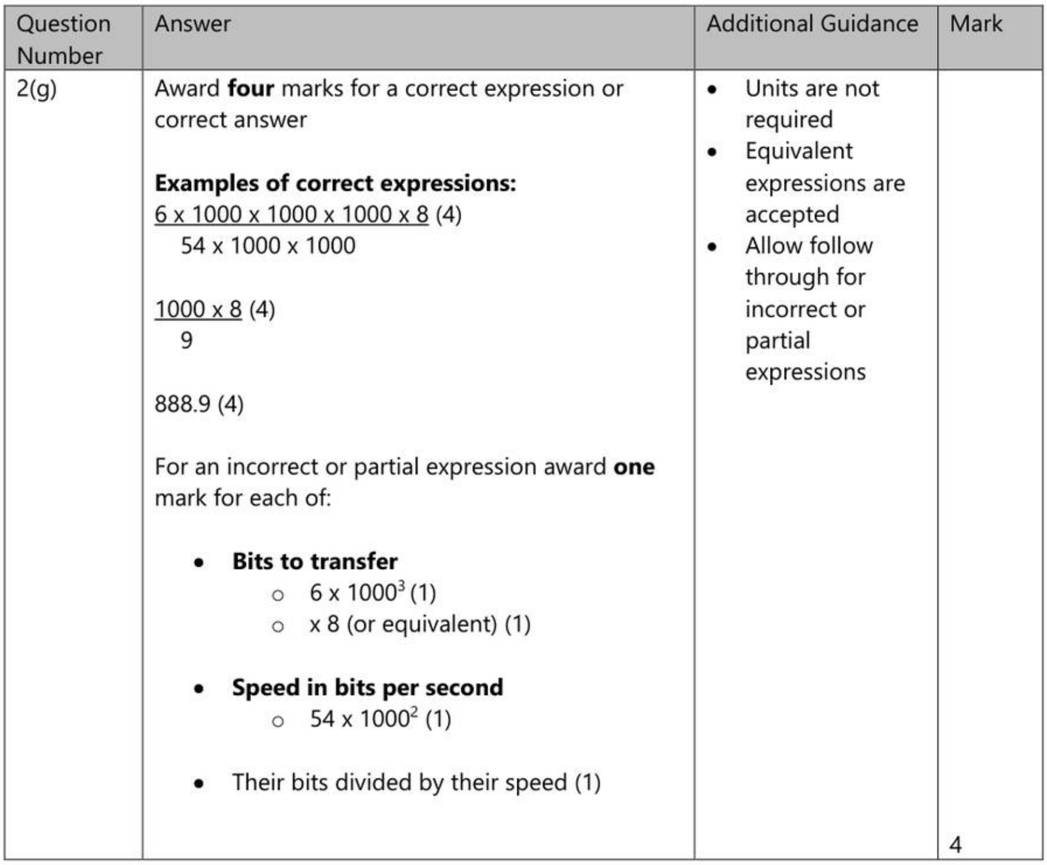
Question Number	Answer	Additional Guidance	Mark
2(e)(i)	Bus		1
2(e)(ii)	Award **one** mark from: • Easy to connect/remove devices/computers (in a network) (1) • Cable cost less / fewer cables needed (compared to mesh/star topology) (1) • Easier to plan/setup/maintain (1)		1
2(e)(iii)	Award **one** mark from: • If the main cable/a terminator fails or gets damaged the whole network will fail (1) • The more workstations the slower it gets (1) • The more workstations the more collisions (1) • Every device ‘sees’ all of the data on the network (1)		1

Question Number	Answer	Additional Guidance	Mark
2(f)(i)	Award **two** marks from: • Reduces time taken to upload/download (1) • Reduces the bandwidth required/cost of data to upload/download (1) • (Usually) produces smaller file/reduces storage (on the web server/device) (1) • People have a limited hearing range / can’t tell the difference (1) • People’s brains will ‘fill in the blanks’ (1)		2
2(f)(ii)	**The only correct answer is C** ***A*** *is not correct because it is not compressed* ***B*** *is not correct because it uses lossless compression* ***D*** *is not correct because it uses lossless compression*		1
2(f)(iii)	Award **one** mark from: • 3c3w2c6w3c (1) • c3w3c2w6c3 (1)		1

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 5 of 10 · Question 2 part a · 1 mark

2021 JUN · Paper 01 · extract 4 of 5 · find part a in this mark scheme extract



Question Number	Answer	Additional Guidance	Mark
2(g)	Award **four** marks for a correct expression or correct answer **Examples of correct expressions:** $\frac{6 \times 1000 \times 1000 \times 1000 \times 8}{54 \times 1000 \times 1000}$ (4) $\frac{1000 \times 8}{9}$ (4) 888.9 (4) For an incorrect or partial expression award **one** mark for each of: • **Bits to transfer** ○ 6×1000^3 (1) ○ x 8 (or equivalent) (1) • **Speed in bits per second** ○ 54×1000^2 (1) • Their bits divided by their speed (1)	• Units are not required • Equivalent expressions are accepted • Allow follow through for incorrect or partial expressions	4

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 5 of 10 · Question 2 part a · 1 mark

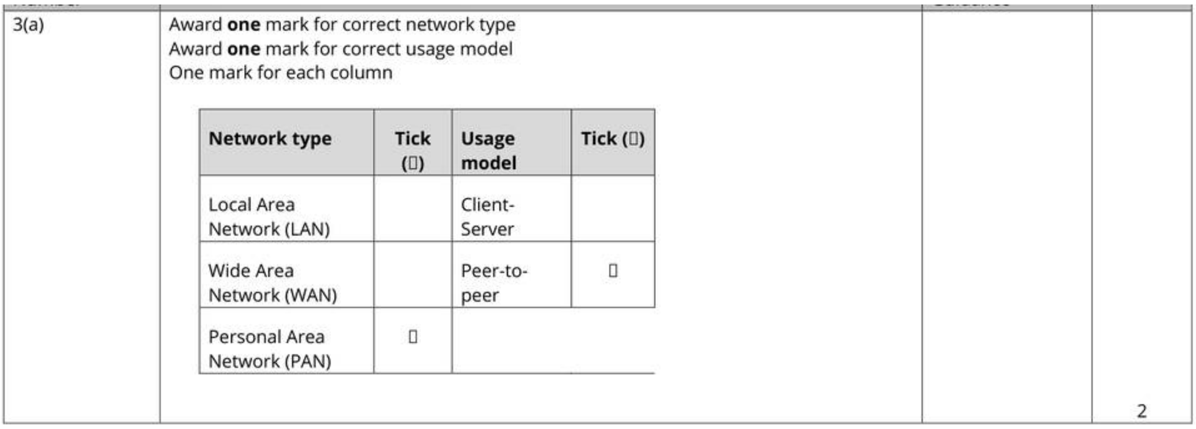
2021 JUN · Paper 01 · extract 5 of 5 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 6 of 10 · Question 3 part a · 2 marks

2021 NOV · Paper 01 · extract 1 of 5 · find part a in this mark scheme extract



3(a)	Award **one** mark for correct network type Award **one** mark for correct usage model One mark for each column		2

Network type	**Tick (▯)**	**Usage model**	**Tick (▯)**
Local Area Network (LAN)		Client-Server	
Wide Area Network (WAN)		Peer-to-peer	▯
Personal Area Network (PAN)	▯		

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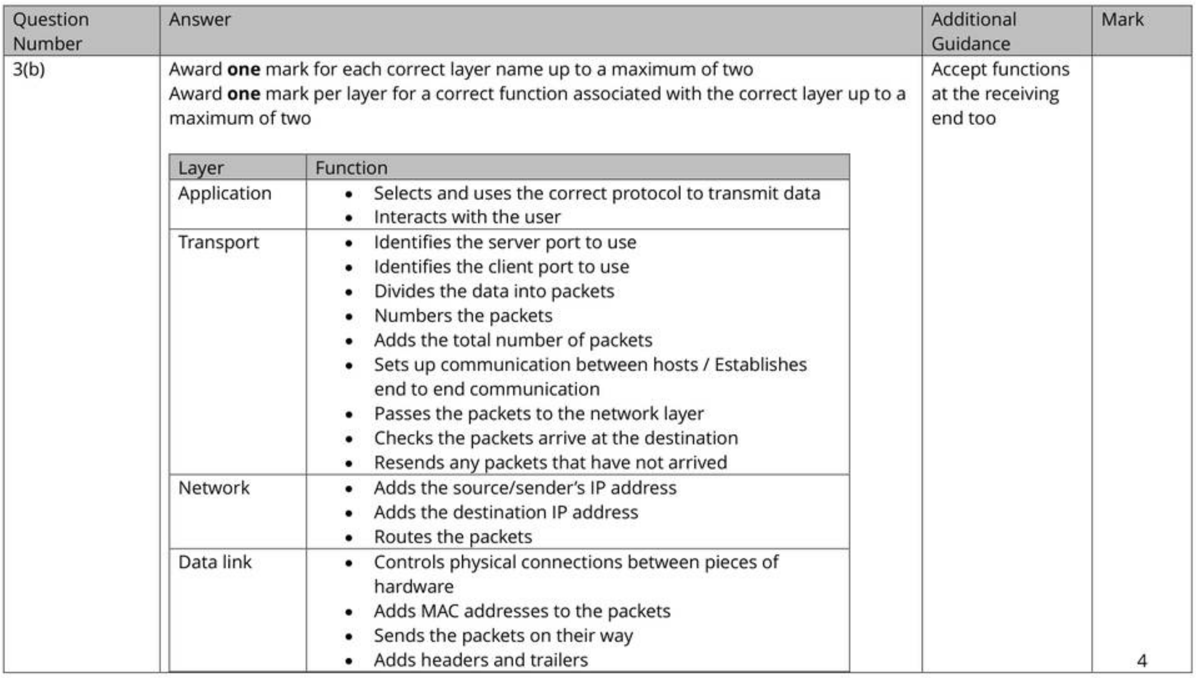

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)</td><td>Award one mark for each correct layer name up to a maximum of two
Award one mark per layer for a correct function associated with the correct layer up to a maximum of two
<table>
<tr><th>Layer</th><th>Function</th></tr>
<tr><td>Application</td><td>• Selects and uses the correct protocol to transmit data
• Interacts with the user</td></tr>
<tr><td>Transport</td><td>• Identifies the server port to use
• Identifies the client port to use
• Divides the data into packets
• Numbers the packets
• Adds the total number of packets
• Sets up communication between hosts / Establishes end to end communication
• Passes the packets to the network layer
• Checks the packets arrive at the destination
• Resends any packets that have not arrived</td></tr>
<tr><td>Network</td><td>• Adds the source/sender's IP address
• Adds the destination IP address
• Routes the packets</td></tr>
<tr><td>Data link</td><td>• Controls physical connections between pieces of hardware
• Adds MAC addresses to the packets
• Sends the packets on their way
• Adds headers and trailers</td></tr>
</table>
</td><td>Accept functions at the receiving end too</td><td>4</td></tr>
</table>

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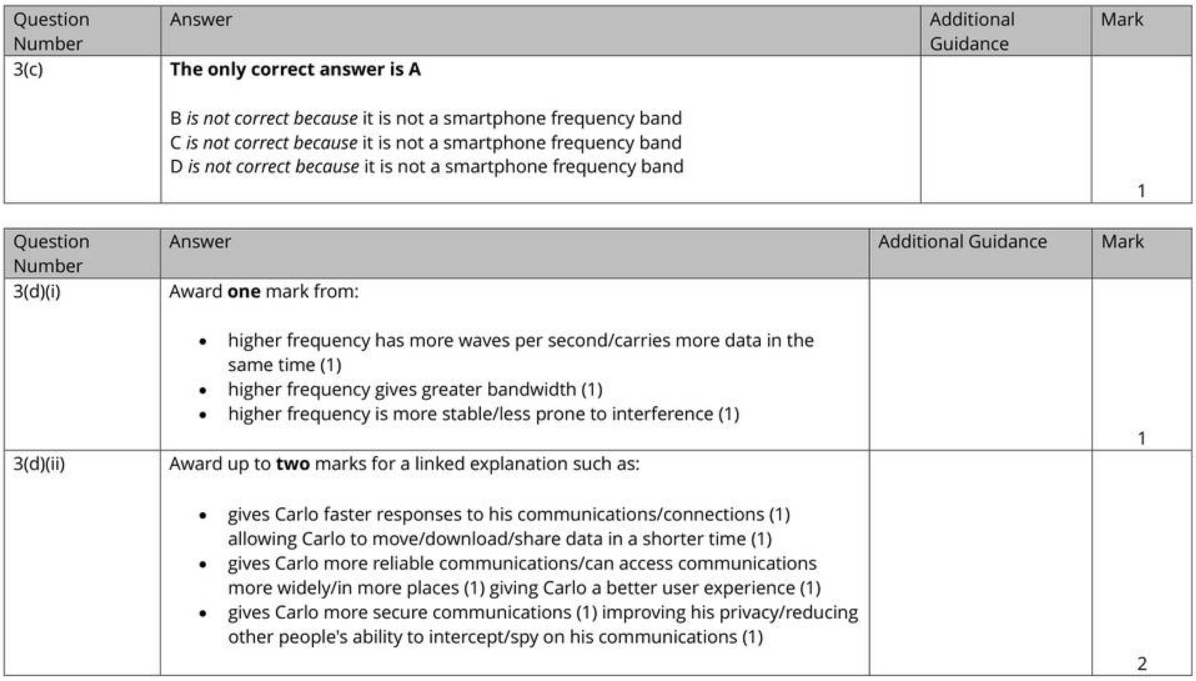

Question Number	Answer	Additional Guidance	Mark
3(c)	**The only correct answer is A** B *is not correct because* it is not a smartphone frequency band C *is not correct because* it is not a smartphone frequency band D *is not correct because* it is not a smartphone frequency band		1

Question Number	Answer	Additional Guidance	Mark
3(d)(i)	Award **one** mark from: • higher frequency has more waves per second/carries more data in the same time (1) • higher frequency gives greater bandwidth (1) • higher frequency is more stable/less prone to interference (1)		1
3(d)(ii)	Award up to **two** marks for a linked explanation such as: • gives Carlo faster responses to his communications/connections (1) allowing Carlo to move/download/share data in a shorter time (1) • gives Carlo more reliable communications/can access communications more widely/in more places (1) giving Carlo a better user experience (1) • gives Carlo more secure communications (1) improving his privacy/reducing other people's ability to intercept/spy on his communications (1)		2

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Answer 6 of 10 · Question 3 part a · 2 marks

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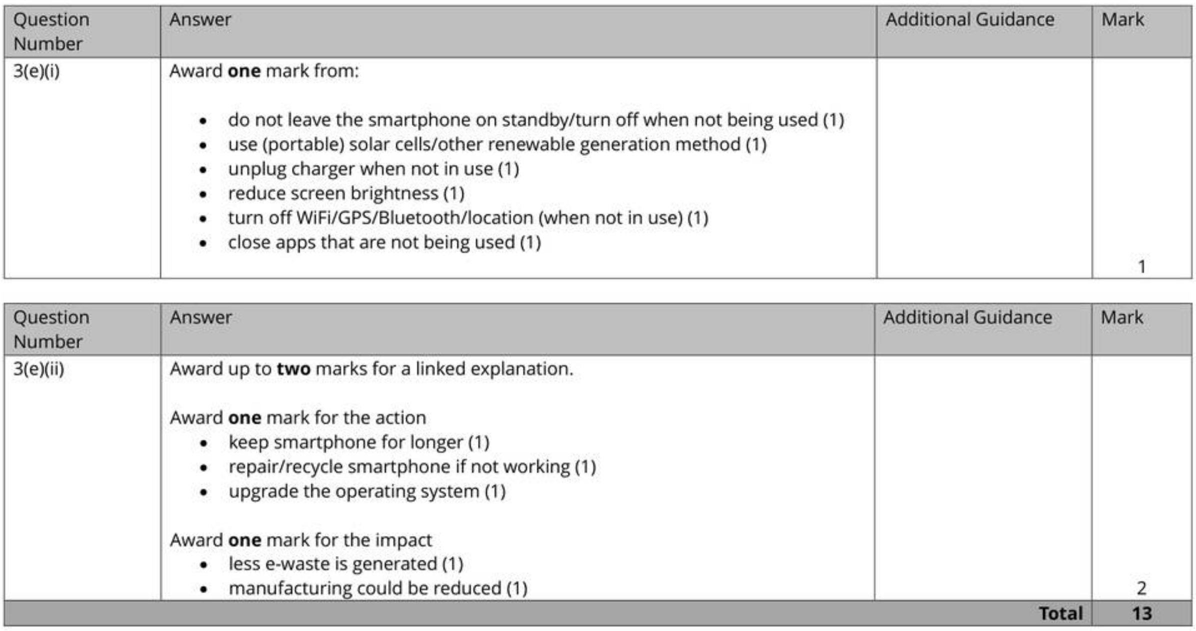


Question Number	Answer	Additional Guidance	Mark
3(e)(i)	Award **one** mark from: • do not leave the smartphone on standby/turn off when not being used (1) • use (portable) solar cells/other renewable generation method (1) • unplug charger when not in use (1) • reduce screen brightness (1) • turn off WiFi/GPS/Bluetooth/location (when not in use) (1) • close apps that are not being used (1)		1

Question Number	Answer	Additional Guidance	Mark
3(e)(ii)	Award up to **two** marks for a linked explanation. Award **one** mark for the action • keep smartphone for longer (1) • repair/recycle smartphone if not working (1) • upgrade the operating system (1) Award **one** mark for the impact • less e-waste is generated (1) • manufacturing could be reduced (1)		2
		Total	**13**

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 6 of 10 · Question 3 part a · 2 marks

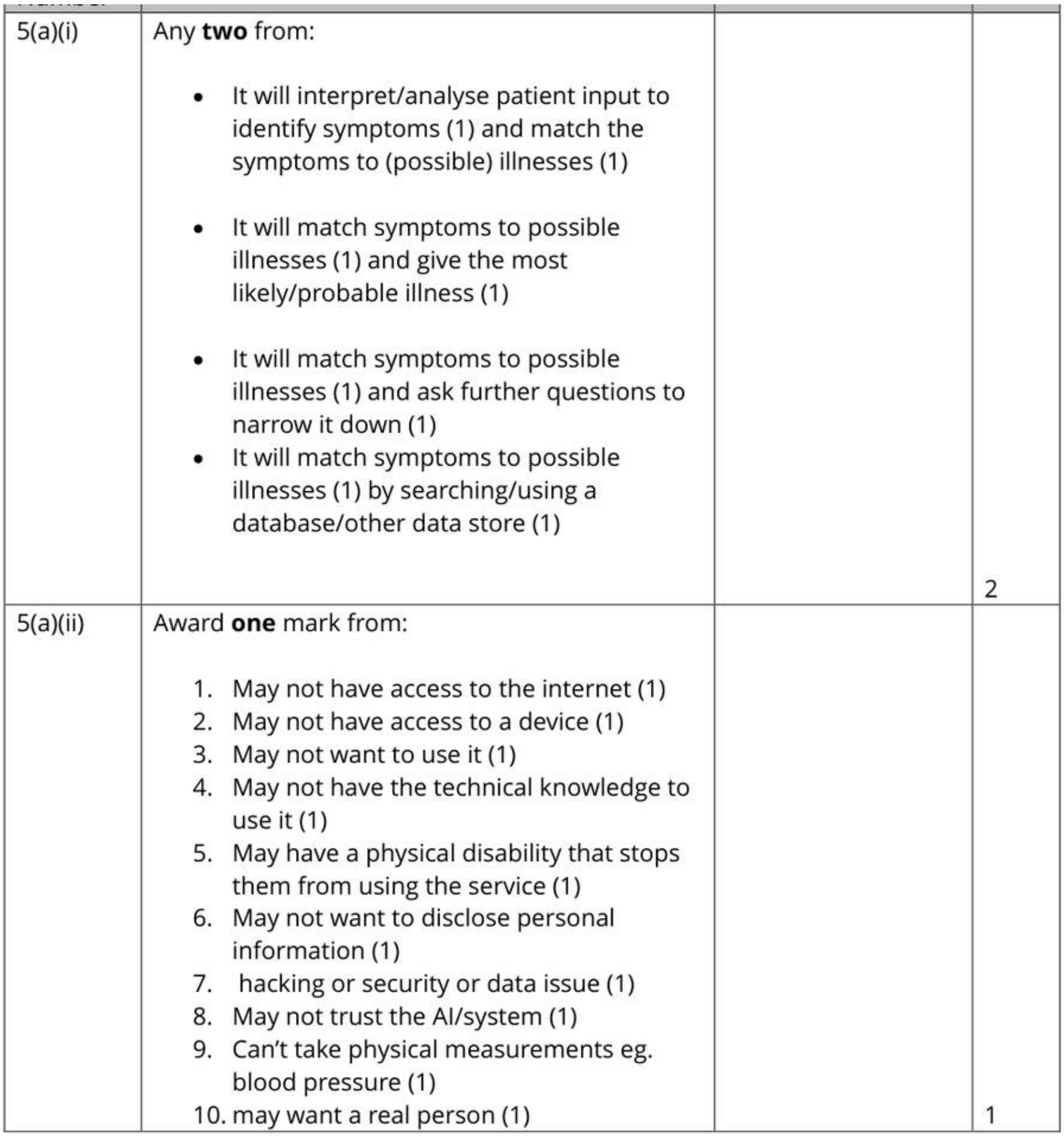
2021 NOV · Paper 01 · extract 5 of 5 · find part a in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 7 of 10 · Question 5 part b.i · 1 mark

2022 JUN · Paper 01 · extract 1 of 4 · find part b.i in this mark scheme extract

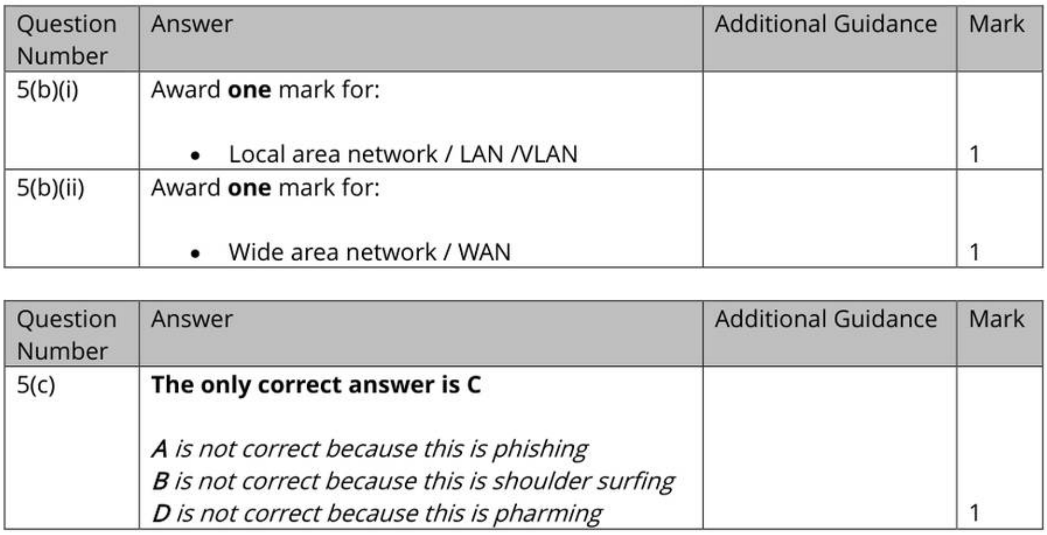


5(a)(i)	Any **two** from: • It will interpret/analyse patient input to identify symptoms (1) and match the symptoms to (possible) illnesses (1) • It will match symptoms to possible illnesses (1) and give the most likely/probable illness (1) • It will match symptoms to possible illnesses (1) and ask further questions to narrow it down (1) • It will match symptoms to possible illnesses (1) by searching/using a database/other data store (1)		2
5(a)(ii)	Award **one** mark from: 1. May not have access to the internet (1) 2. May not have access to a device (1) 3. May not want to use it (1) 4. May not have the technical knowledge to use it (1) 5. May have a physical disability that stops them from using the service (1) 6. May not want to disclose personal information (1) 7. hacking or security or data issue (1) 8. May not trust the AI/system (1) 9. Can't take physical measurements eg. blood pressure (1) 10. may want a real person (1)		1

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 7 of 10 · Question 5 part b.i · 1 mark

2022 JUN · Paper 01 · extract 2 of 4 · find part b.i in this mark scheme extract



Question Number	Answer	Additional Guidance	Mark
5(b)(i)	Award **one** mark for: • Local area network / LAN /VLAN		1
5(b)(ii)	Award **one** mark for: • Wide area network / WAN		1

Question Number	Answer	Additional Guidance	Mark
5(c)	**The only correct answer is C** *A is not correct because this is phishing* *B is not correct because this is shoulder surfing* *D is not correct because this is pharming*		1

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 7 of 10 · Question 5 part b.i · 1 mark

2022 JUN · Paper 01 · extract 3 of 4 · find part b.i in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(i)</td><td>Award up to two marks for a linked explanation such as:
Robust/less likely to break it if dropped / quieter than a mechanical hard drive (1) because it doesn’t have any moving/mechanical parts (1)
Smaller/thinner/lighter than a device with HDD (1) so easier to carry/more portable / laptops have less room in them for drives/componants(1)
SSD uses less power (1) so battery will last longer (1)
SSD is faster (1) so can access data/records more quickly (1)</td><td></td><td>2</td></tr>
<tr><td>5(d)(ii)</td><td>Any two from:
Flash memory chips are used (1)
Chips have (floating gate) transistors/electron pools/charge traps (1)
Gates/electron pools/charge traps hold an electrical charge (1)
Charge remains even when no power (1)
No charge/empty pool/trap represents data/1 (1)
A charge/full pool/trap represents no data/0 (1)
Data is stored in blocks and pages (1)</td><td></td><td>2</td></tr>
<tr><td>5(d)(iii)</td><td>Award one mark for each of:<table><tr><th>Description</th><th>RAM</th><th>ROM</th></tr><tr><td>Stores the boot up sequence</td><td></td><td>▯</td></tr><tr><td>The contents are lost when the laptop is shut down</td><td>▯</td><td></td></tr></table></td><td></td><td>2</td></tr>
<tr><td colspan="3" align="right">Total for question 5</td><td>12</td></tr>
</table>

Answer 7 of 10 · Question 5 part b.i · 1 mark

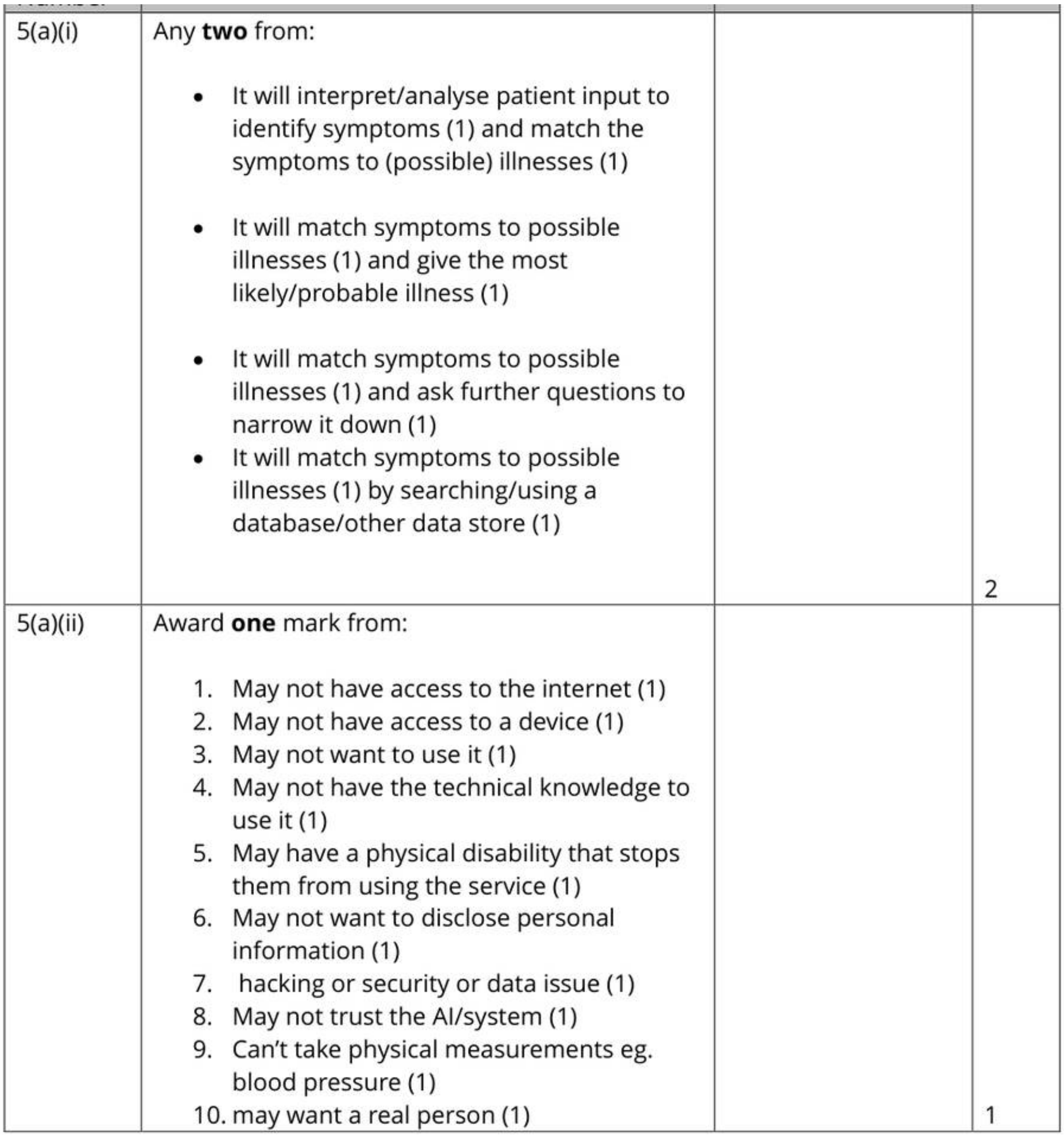
2022 JUN · Paper 01 · extract 4 of 4 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 8 of 10 · Question 5 part b.ii · 1 mark

2022 JUN · Paper 01 · extract 1 of 4 · find part b.ii in this mark scheme extract

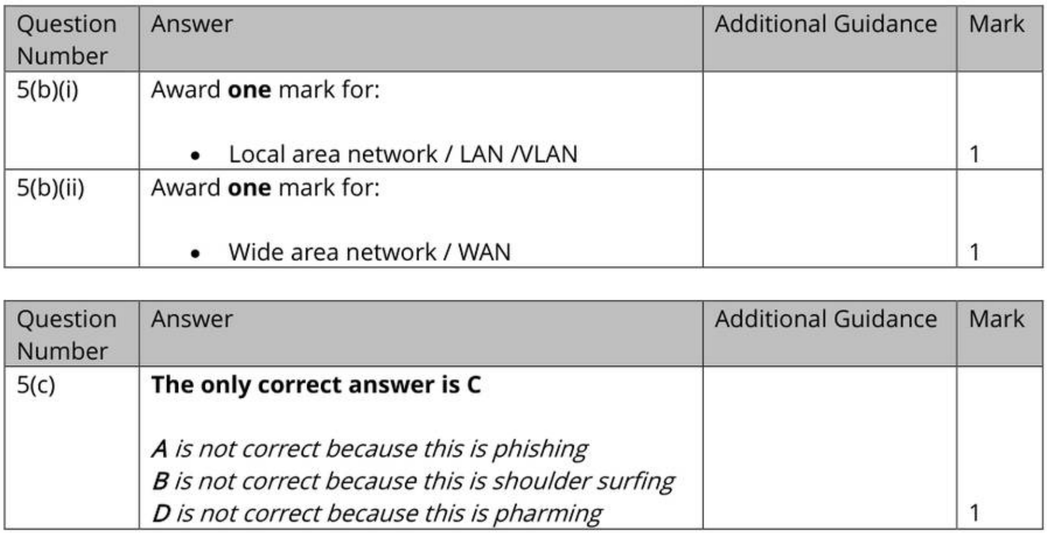


5(a)(i)	Any **two** from: • It will interpret/analyse patient input to identify symptoms (1) and match the symptoms to (possible) illnesses (1) • It will match symptoms to possible illnesses (1) and give the most likely/probable illness (1) • It will match symptoms to possible illnesses (1) and ask further questions to narrow it down (1) • It will match symptoms to possible illnesses (1) by searching/using a database/other data store (1)		2
5(a)(ii)	Award **one** mark from: 1. May not have access to the internet (1) 2. May not have access to a device (1) 3. May not want to use it (1) 4. May not have the technical knowledge to use it (1) 5. May have a physical disability that stops them from using the service (1) 6. May not want to disclose personal information (1) 7. hacking or security or data issue (1) 8. May not trust the AI/system (1) 9. Can't take physical measurements eg. blood pressure (1) 10. may want a real person (1)		1

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 8 of 10 · Question 5 part b.ii · 1 mark

2022 JUN · Paper 01 · extract 2 of 4 · find part b.ii in this mark scheme extract



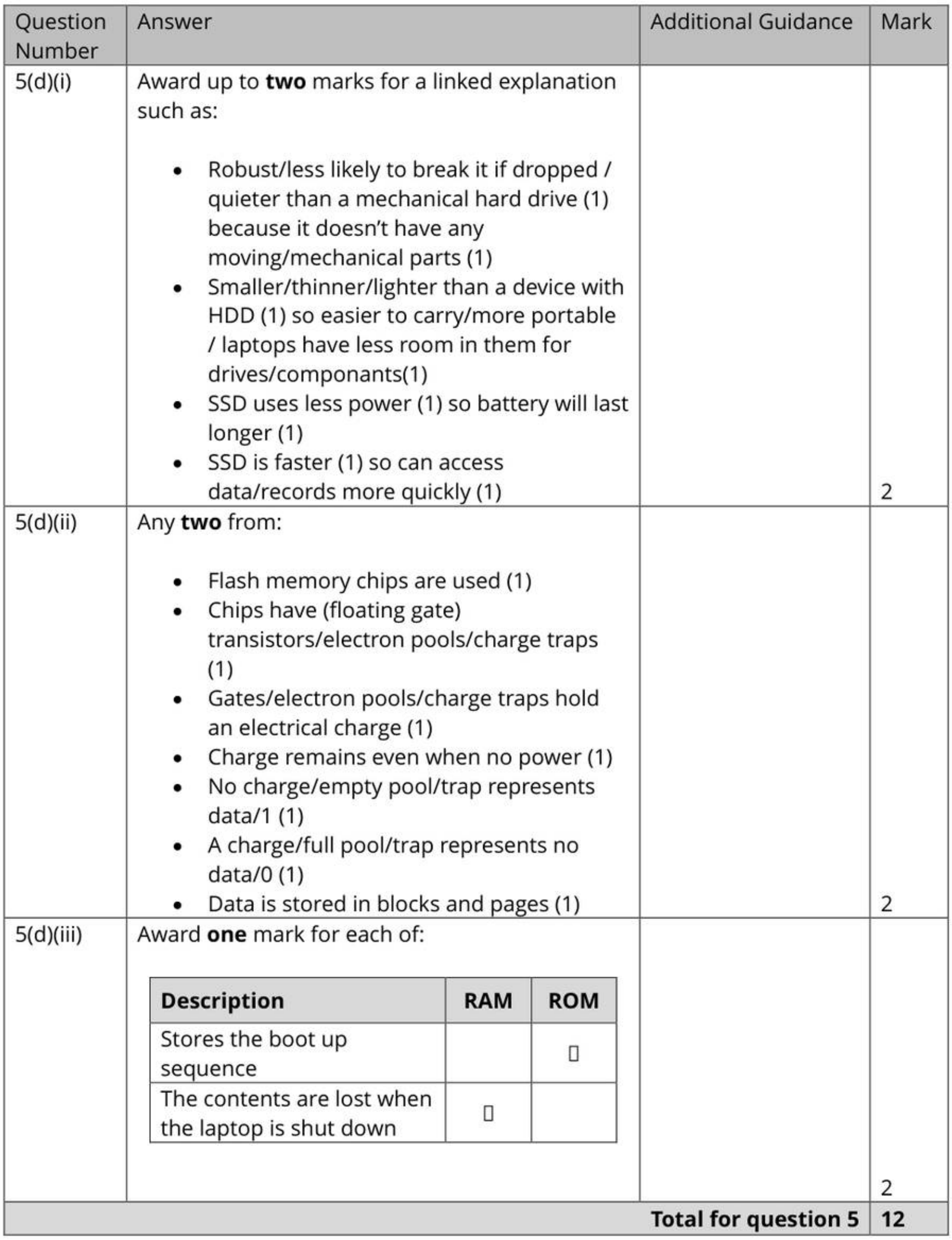
Question Number	Answer	Additional Guidance	Mark
5(b)(i)	Award **one** mark for: - Local area network / LAN /VLAN		1
5(b)(ii)	Award **one** mark for: - Wide area network / WAN		1

Question Number	Answer	Additional Guidance	Mark
5(c)	**The only correct answer is C** *A is not correct because this is phishing* *B is not correct because this is shoulder surfing* *D is not correct because this is pharming*		1

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 8 of 10 · Question 5 part b.ii · 1 mark

2022 JUN · Paper 01 · extract 3 of 4 · find part b.ii in this mark scheme extract



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<tr><td>5(d)(ii)</td><td>Any two from:
• Flash memory chips are used (1)
• Chips have (floating gate) transistors/electron pools/charge traps (1)
• Gates/electron pools/charge traps hold an electrical charge (1)
• Charge remains even when no power (1)
• No charge/empty pool/trap represents data/1 (1)
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</table>
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</table>

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 8 of 10 · Question 5 part b.ii · 1 mark

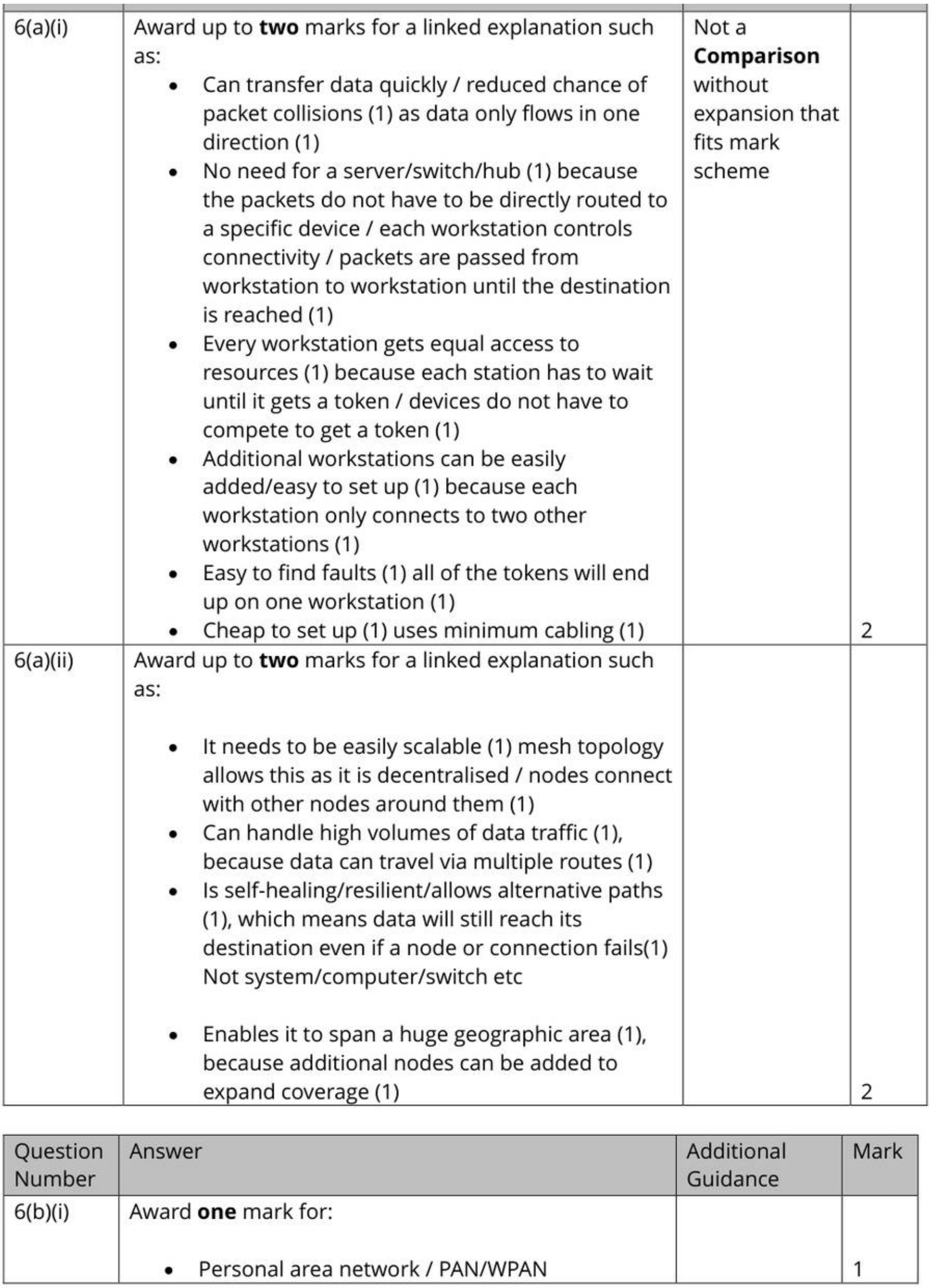
2022 JUN · Paper 01 · extract 4 of 4 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 9 of 10 · Question 6 part b.i · 1 mark

2022 JUN · Paper 01 · extract 1 of 4 · find part b.i in this mark scheme extract



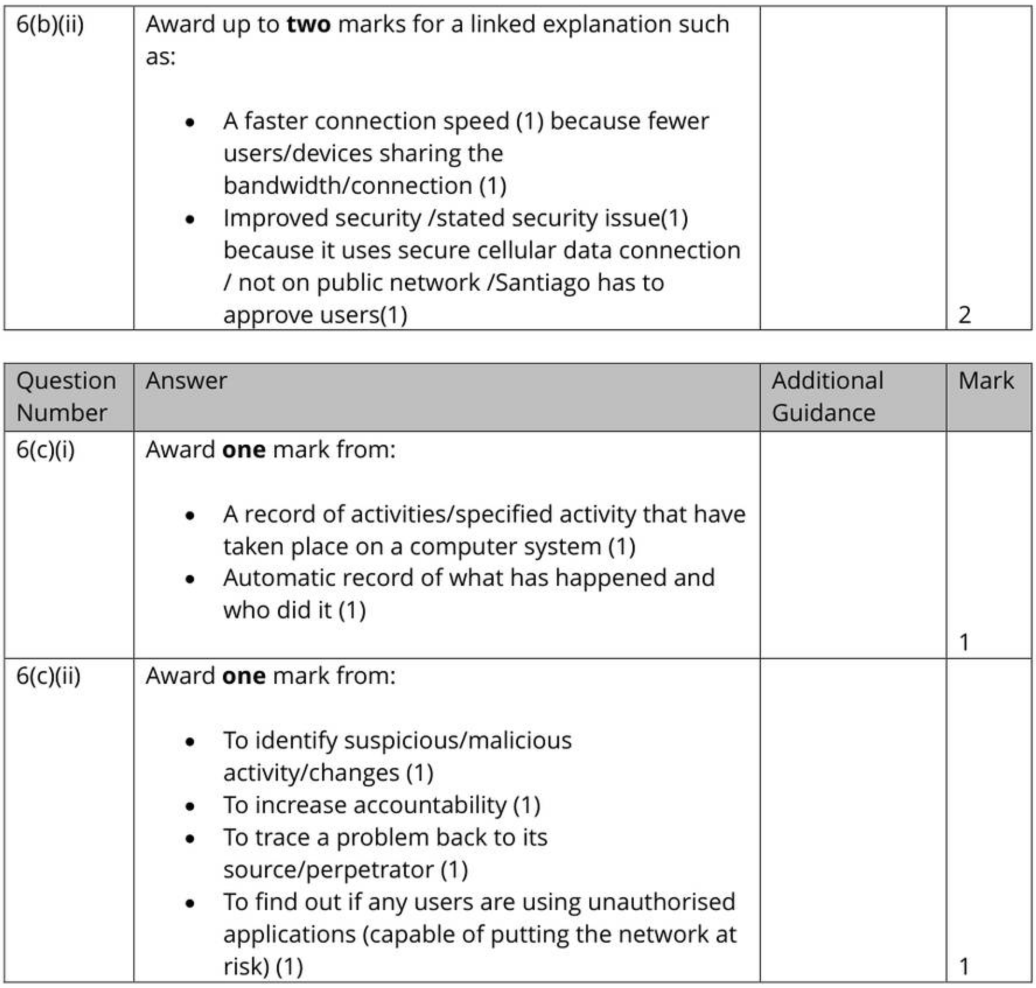
6(a)(i)	Award up to **two** marks for a linked explanation such as: • Can transfer data quickly / reduced chance of packet collisions (1) as data only flows in one direction (1) • No need for a server/switch/hub (1) because the packets do not have to be directly routed to a specific device / each workstation controls connectivity / packets are passed from workstation to workstation until the destination is reached (1) • Every workstation gets equal access to resources (1) because each station has to wait until it gets a token / devices do not have to compete to get a token (1) • Additional workstations can be easily added/easy to set up (1) because each workstation only connects to two other workstations (1) • Easy to find faults (1) all of the tokens will end up on one workstation (1) • Cheap to set up (1) uses minimum cabling (1)	Not a **Comparison** without expansion that fits mark scheme	2
6(a)(ii)	Award up to **two** marks for a linked explanation such as: • It needs to be easily scalable (1) mesh topology allows this as it is decentralised / nodes connect with other nodes around them (1) • Can handle high volumes of data traffic (1), because data can travel via multiple routes (1) • Is self-healing/resilient/allows alternative paths (1), which means data will still reach its destination even if a node or connection fails(1) Not system/computer/switch etc • Enables it to span a huge geographic area (1), because additional nodes can be added to expand coverage (1)		2

Question Number	Answer	Additional Guidance	Mark
6(b)(i)	Award **one** mark for: • Personal area network / PAN/WPAN		1

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 9 of 10 · Question 6 part b.i · 1 mark

2022 JUN · Paper 01 · extract 2 of 4 · find part b.i in this mark scheme extract



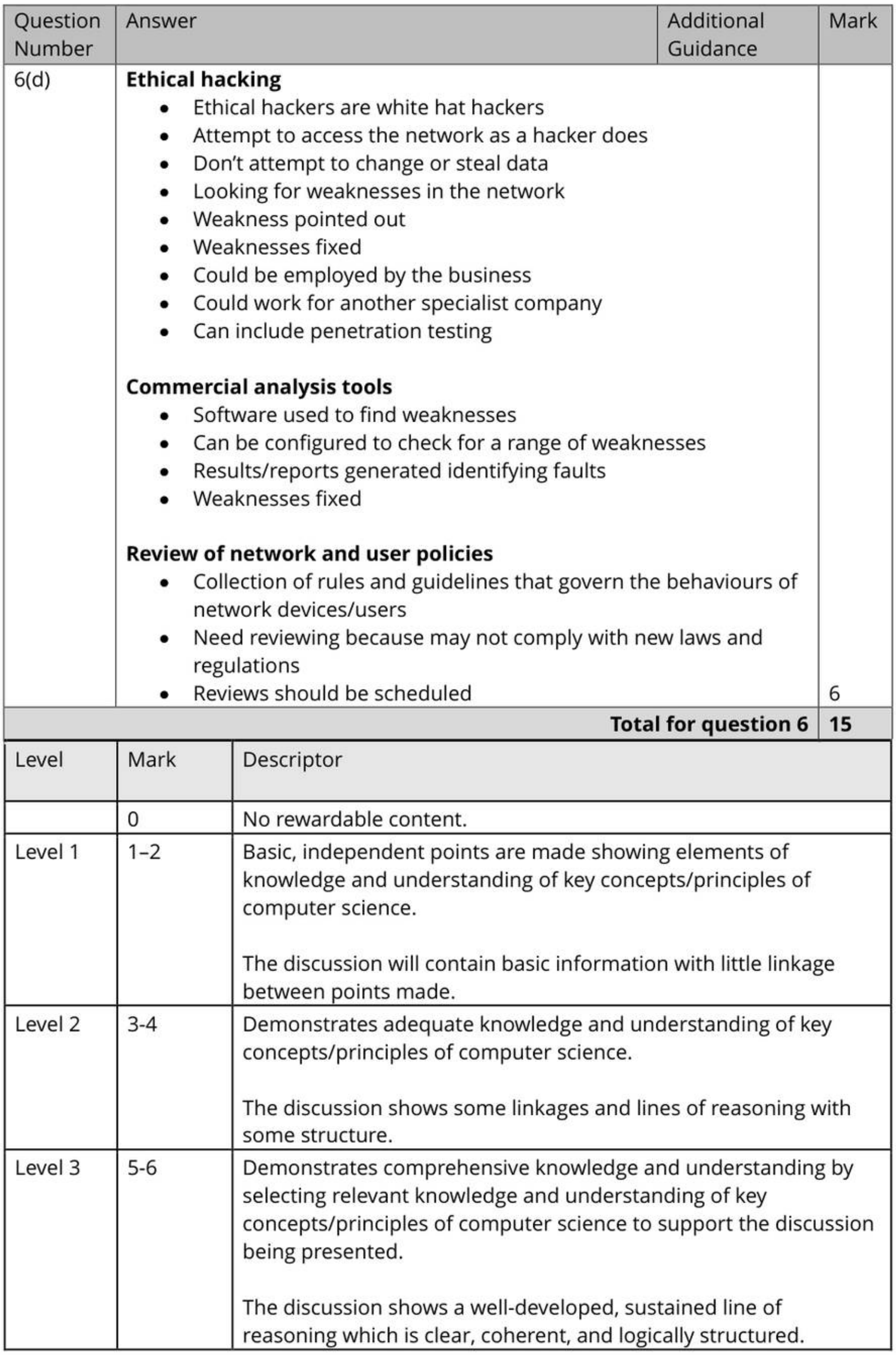
6(b)(ii)	Award up to **two** marks for a linked explanation such as: - A faster connection speed (1) because fewer users/devices sharing the bandwidth/connection (1) - Improved security /stated security issue(1) because it uses secure cellular data connection / not on public network /Santiago has to approve users(1)		2

Question Number	Answer	Additional Guidance	Mark
6(c)(i)	Award **one** mark from: - A record of activities/specified activity that have taken place on a computer system (1) - Automatic record of what has happened and who did it (1)		1
6(c)(ii)	Award **one** mark from: - To identify suspicious/malicious activity/changes (1) - To increase accountability (1) - To trace a problem back to its source/perpetrator (1) - To find out if any users are using unauthorised applications (capable of putting the network at risk) (1)		1

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 9 of 10 · Question 6 part b.i · 1 mark

2022 JUN · Paper 01 · extract 3 of 4 · find part b.i in this mark scheme extract



Question Number	Answer	Additional Guidance	Mark
6(d)	**Ethical hacking** - Ethical hackers are white hat hackers - Attempt to access the network as a hacker does - Don't attempt to change or steal data - Looking for weaknesses in the network - Weakness pointed out - Weaknesses fixed - Could be employed by the business - Could work for another specialist company - Can include penetration testing **Commercial analysis tools** - Software used to find weaknesses - Can be configured to check for a range of weaknesses - Results/reports generated identifying faults - Weaknesses fixed **Review of network and user policies** - Collection of rules and guidelines that govern the behaviours of network devices/users - Need reviewing because may not comply with new laws and regulations - Reviews should be scheduled		6
		Total for question 6	**15**

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

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Type MS · Network types: LAN, WAN, PAN and why we network

Answer 9 of 10 · Question 6 part b.i · 1 mark

2022 JUN · Paper 01 · extract 4 of 4 · find part b.i in this mark scheme extract

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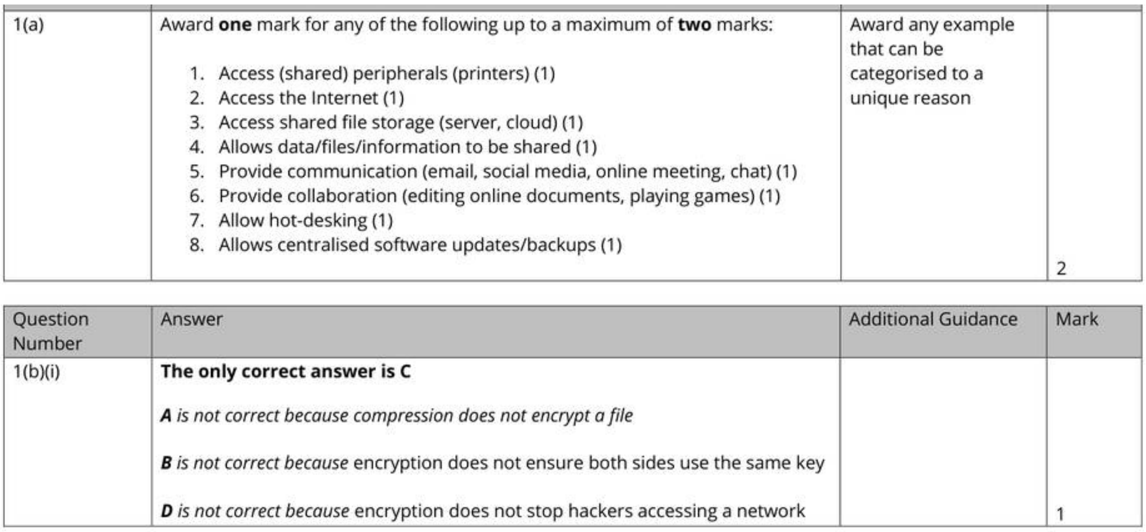
Type MS · Network types: LAN, WAN, PAN and why we network

Answer 10 of 10 · Question 1 part a · 2 marks

2023 JUN · Paper 01 · extract 1 of 4 · find part a in this mark scheme extract



1(a)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. Access (shared) peripherals (printers) (1) 2. Access the Internet (1) 3. Access shared file storage (server, cloud) (1) 4. Allows data/files/information to be shared (1) 5. Provide communication (email, social media, online meeting, chat) (1) 6. Provide collaboration (editing online documents, playing games) (1) 7. Allow hot-desking (1) 8. Allows centralised software updates/backups (1)	Award any example that can be categorised to a unique reason	2

Question Number	Answer	Additional Guidance	Mark
1(b)(i)	**The only correct answer is C** ***A*** *is not correct because compression does not encrypt a file* ***B*** *is not correct because* encryption does not ensure both sides use the same key ***D*** *is not correct because* encryption does not stop hackers accessing a network		1

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 10 of 10 · Question 1 part a · 2 marks

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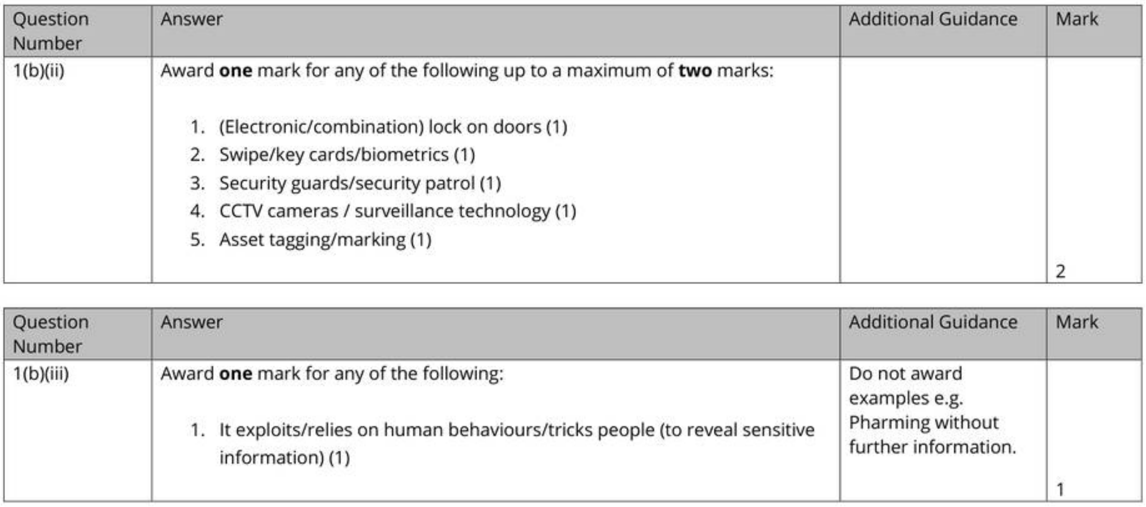

Question Number	Answer	Additional Guidance	Mark
1(b)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. (Electronic/combination) lock on doors (1) 2. Swipe/key cards/biometrics (1) 3. Security guards/security patrol (1) 4. CCTV cameras / surveillance technology (1) 5. Asset tagging/marking (1)		2

Question Number	Answer	Additional Guidance	Mark
1(b)(iii)	Award **one** mark for any of the following: 1. It exploits/relies on human behaviours/tricks people (to reveal sensitive information) (1)	Do not award examples e.g. Pharming without further information.	1

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 10 of 10 · Question 1 part a · 2 marks

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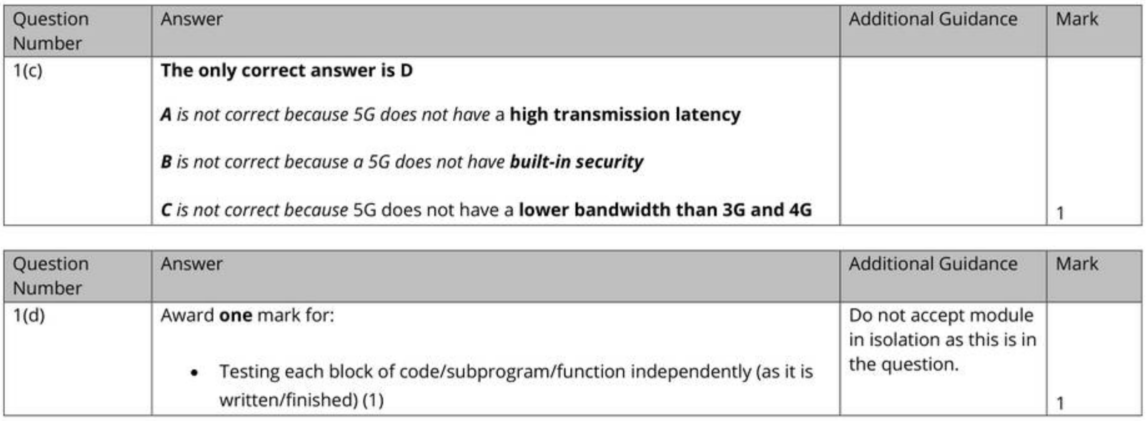

Question Number	Answer	Additional Guidance	Mark
1(c)	**The only correct answer is D** ***A** is not correct because 5G does not have* a **high transmission latency** ***B** is not correct because a 5G does not have* ***built-in security*** ***C** is not correct because* 5G does not have a **lower bandwidth than 3G and 4G**		1

Question Number	Answer	Additional Guidance	Mark
1(d)	Award **one** mark for: • Testing each block of code/subprogram/function independently (as it is written/finished) (1)	Do not accept module in isolation as this is in the question.	1

Type MS · Network types: LAN, WAN, PAN and why we network

Answer 10 of 10 · Question 1 part a · 2 marks

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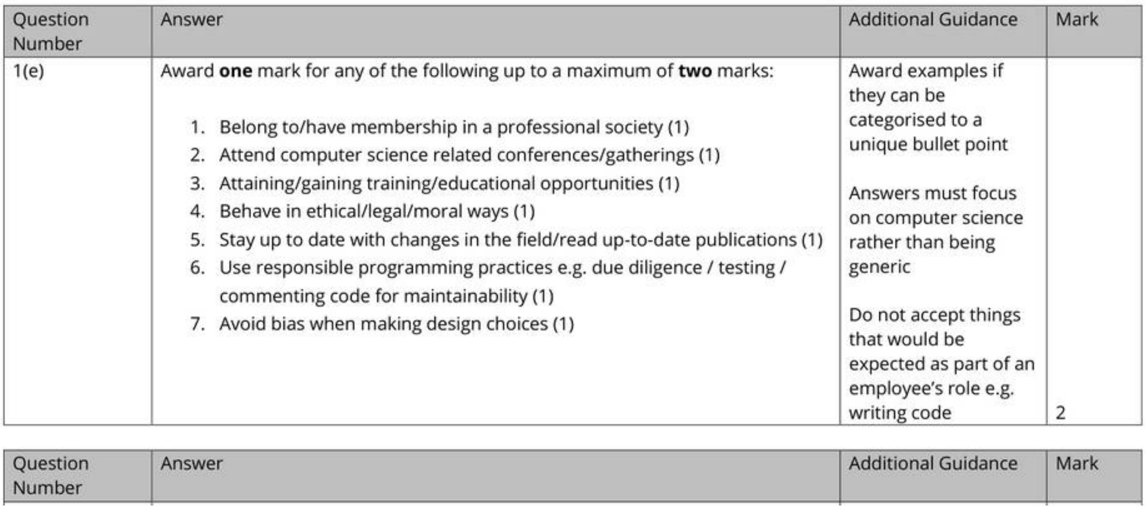

Question Number	Answer	Additional Guidance	Mark
1(e)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. Belong to/have membership in a professional society (1) 2. Attend computer science related conferences/gatherings (1) 3. Attaining/gaining training/educational opportunities (1) 4. Behave in ethical/legal/moral ways (1) 5. Stay up to date with changes in the field/read up-to-date publications (1) 6. Use responsible programming practices e.g. due diligence / testing / commenting code for maintainability (1) 7. Avoid bias when making design choices (1)	Award examples if they can be categorised to a unique bullet point Answers must focus on computer science rather than being generic Do not accept things that would be expected as part of an employee's role e.g. writing code	2

Question Number	Answer	Additional Guidance	Mark